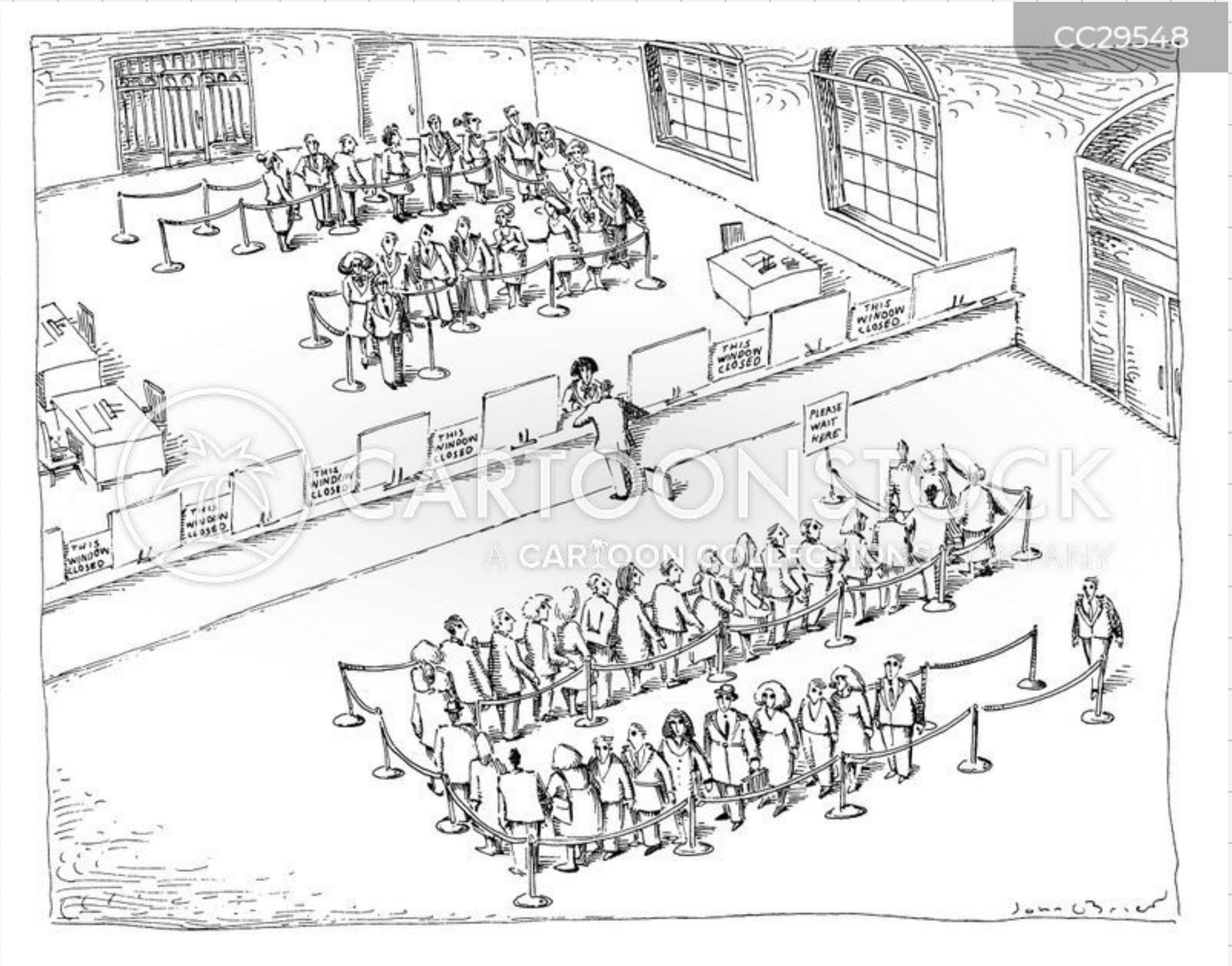
Spring 2025

IS 365

Topic:

Process Analysis (Quantitative)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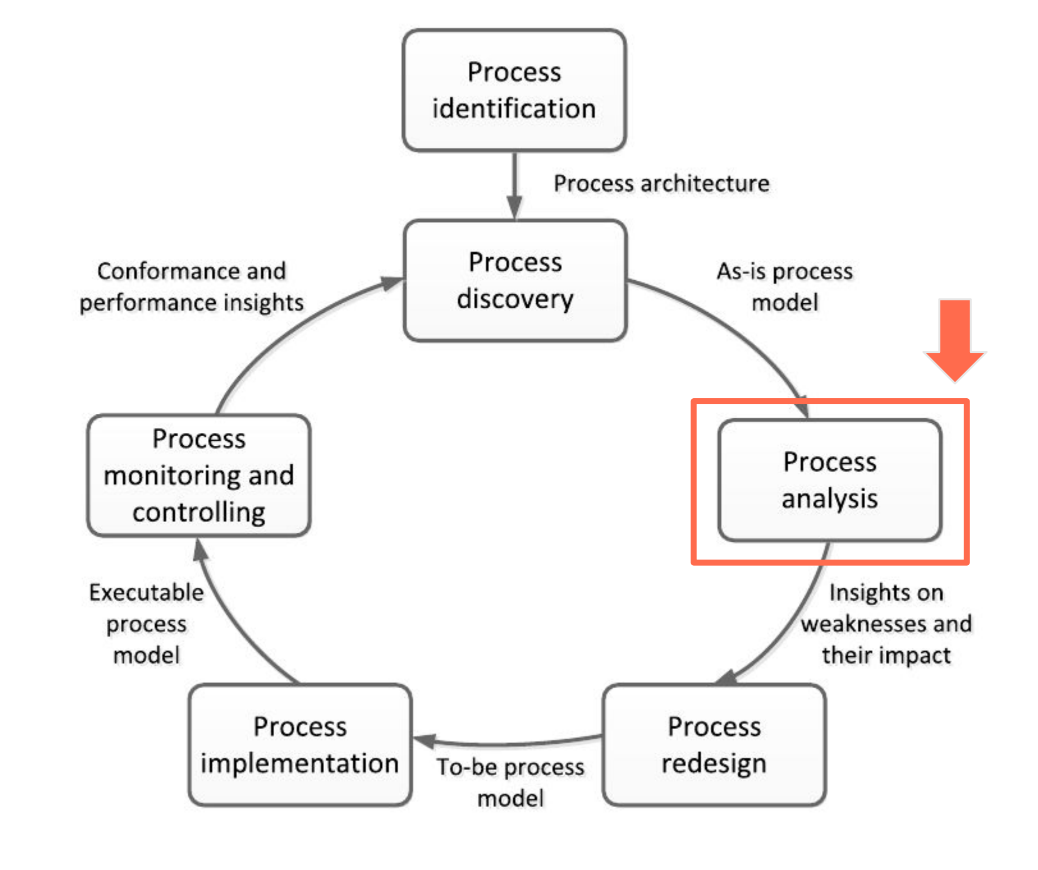
CARTOONSTOCK
A CARTOON COLLECTIONS COMPANY

Announcements

- Assignment #2 posted; due March 16

Recap

Focus Area



Process identification
Process architecture
Process discovery
As-is process model
Process analysis
Insights on weaknesses and their impact
Process redesign
To-be process model
Process implementation
Executable process model
Process monitoring and controlling
Conformance and performance insights

Value-Added Analysis

- Identify unnecessary or redundant steps in processes
- **Separate tasks into:** Value Adding (VA), Business Value Adding (BVA), Non-Value Adding (NVA)

Waste Analysis

- Eliminate or minimize waste categories
- Align process with customer value
- **Types of waste:** move, hold, overdo

Issue Register

- **Structured list of issues with fields:** description, priority, data assumptions
- Prioritize problems using Pareto or CHIP charts

Root Cause Analysis

- Avoids treating symptoms; focuses on underlying problems
- Helps in designing sustainable process improvements
- **Methods:** Ishikawa diagram, why-why analysis

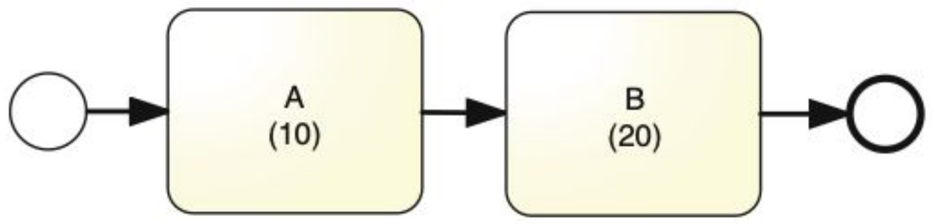
Flow Analysis

Purpose of Flow Analysis

- **Estimate process performance:** cycle time, cost, error rates
- Identify bottlenecks and improvement opportunities

Sequential

Sequential: Example



Sequential: Example

A loan application has two sequential steps:

Verify Documents	**Credit Check**
T_1 = 10 min	T_2 = 20 min

Sequential: Analysis

A loan application has two sequential steps:

Verify Documents

$T_1 = 10$ min

Credit Check

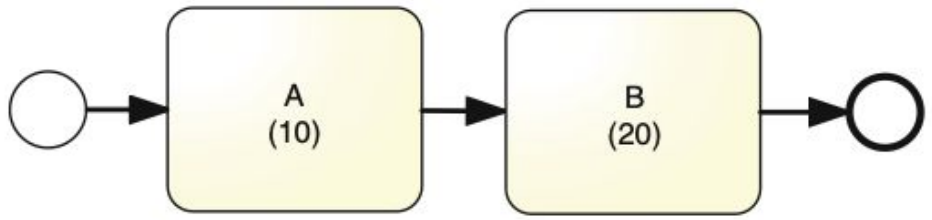
$T_2 = 20$ min

$$CT = T_1 + T_2 = 10 + 20$$

CT = 30 min

Each step must complete before the next begins — just add all task times.

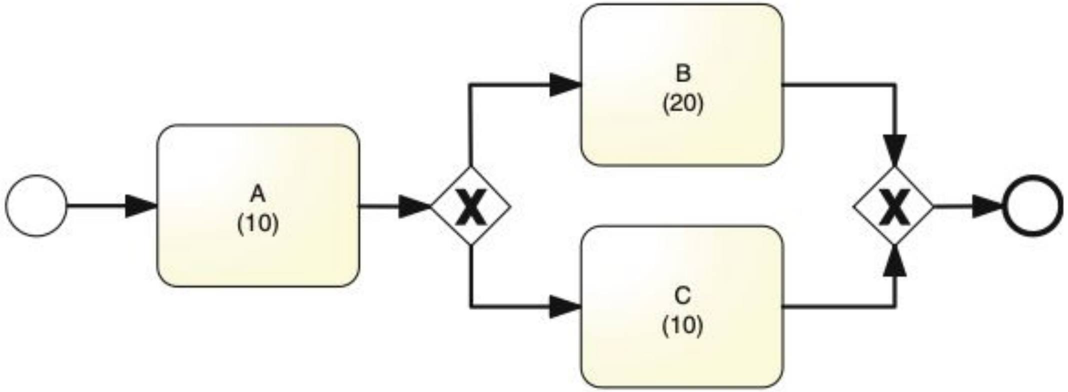
Sequential: Generalization



$$CT = \sum_{i=1}^{n} T_i$$

XOR Gateway

XOR Gateway: Example



XOR Gateway: Example

After intake, an insurance claim goes to one of two paths (not both):

Complex Review	Simple Review
$p = 0.3$ | $T = 20$ min	$p = 0.7$ | $T = 10$ min

XOR Gateway: Example

After intake, an insurance claim goes to one of two paths (not both):

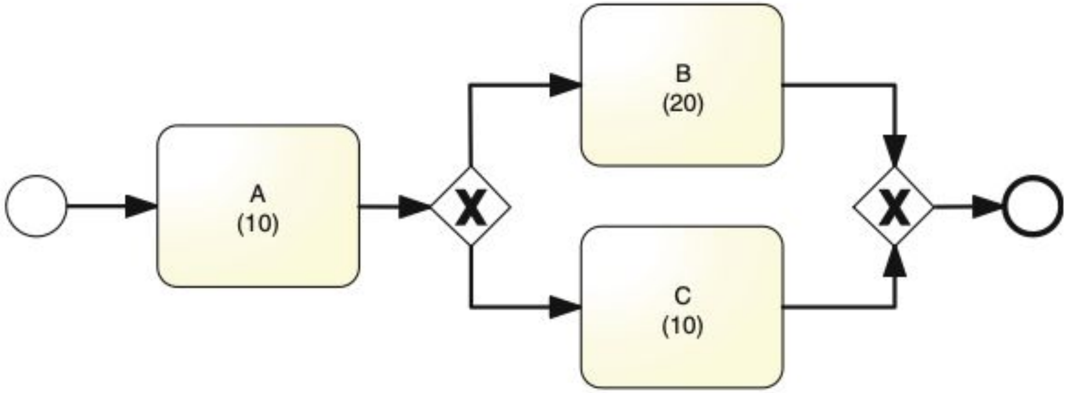
Complex Review	**Simple Review**
$p = 0.3$ | $T = 20$ min	$p = 0.7$ | $T = 10$ min
Contribution: 0.3 × 20 = 6 min	**Contribution: 0.7 × 10 = 7 min**

$$CT = \sum p_i \times T_i = 6 + 7$$

CT = 13 min

Only ONE path executes per instance — weight each branch's time by its probability.

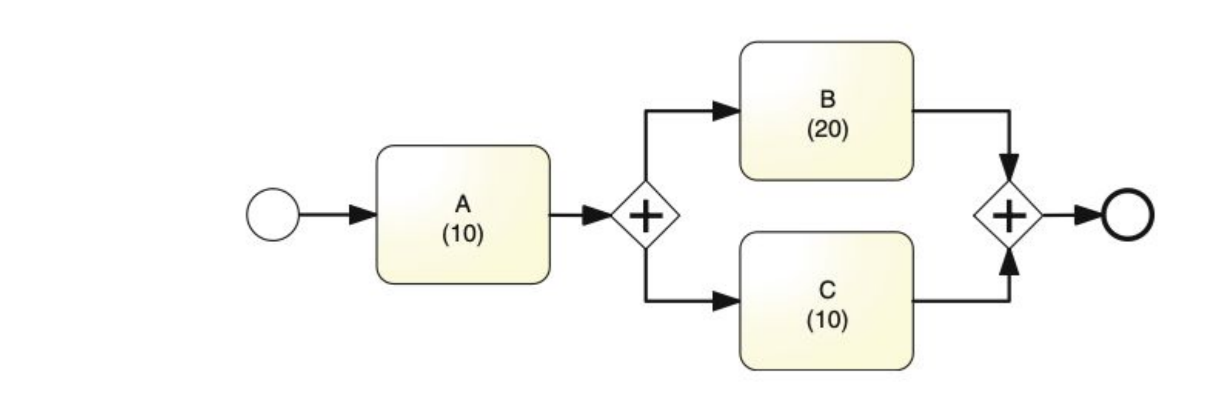
XOR Gateway: Generalization



$$CT = \sum_{i=1}^{n} p_i \times T_i$$

AND Gateway

AND Gateway: Example



AND Gateway: Example

Order fulfillment runs two tasks in parallel (both must finish):

Pack Items	**Print Label**
$T_1 = 20$ min	$T_2 = 10$ min

AND Gateway: Example

Order fulfillment runs two tasks in parallel (both must finish):

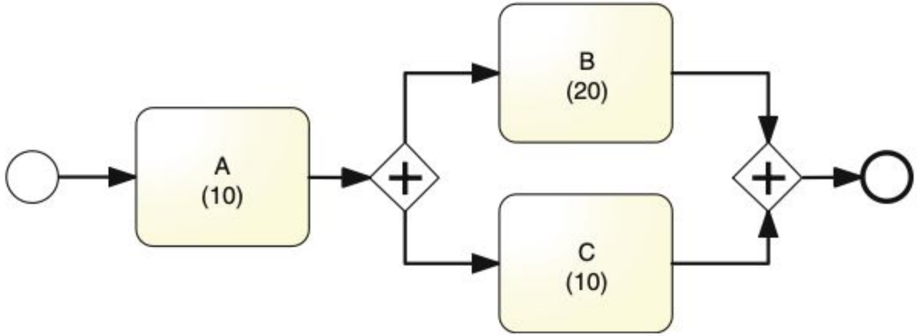
Pack Items	Print Label
$T_1 = 20$ min ← longest	$T_2 = 10$ min

$$CT = \text{Max}(T_1, T_2) = \text{Max}(20, 10)$$

CT = 20 min

ALL branches run simultaneously — the process waits for the slowest branch.

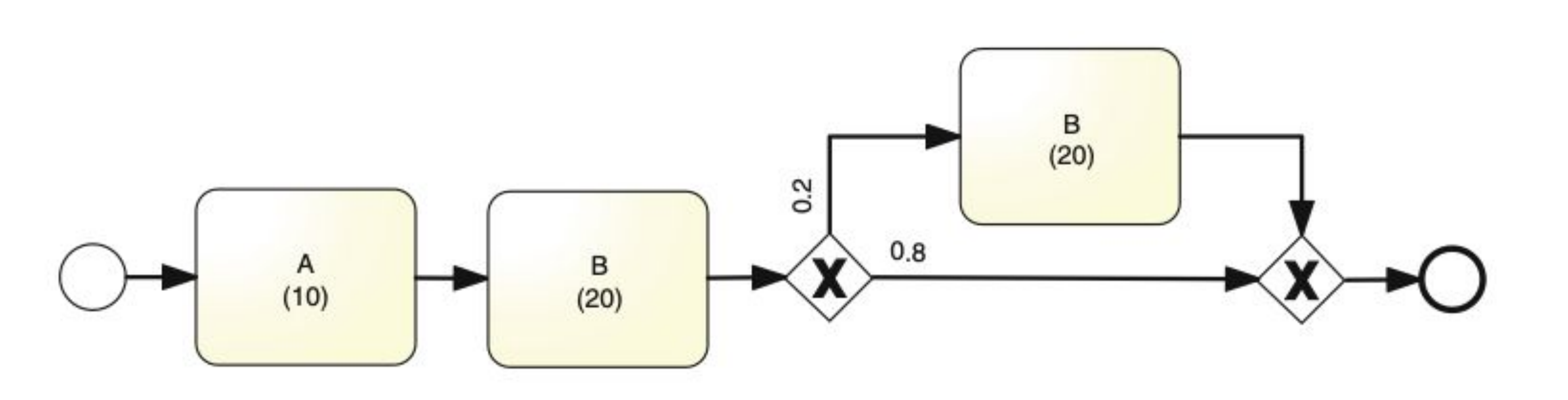
AND Gateway: Generalization



$$CT = Max(T_1, T_2, \ldots, T_n)$$

Rework Loop

Rework Loop: Example



Rework Loop: Example

A quality check sends 20% of reports back for revision:

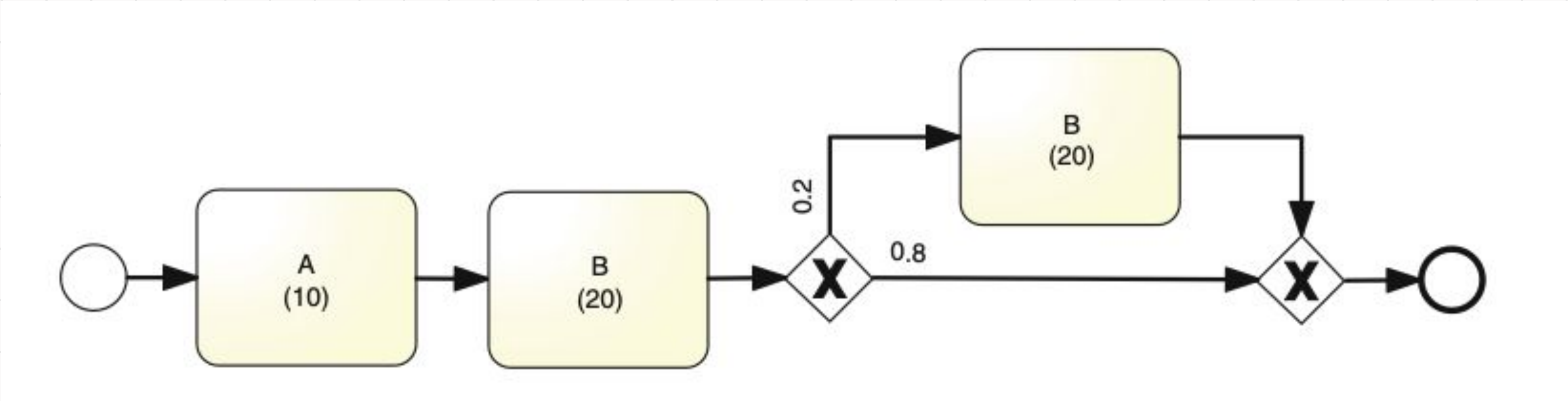
Task time (T)	**20 min**
Rework rate (r)	**0.2 (20%)**
Pass rate (1 – r)	**0.8 (80%)**

$$CT = T / (1 - r) = 20 / 0.8$$

CT = 25 min

Rework inflates average time: even rare loops add up over many instances.

Rework Loop: Generalization



$$CT = \frac{T}{1 - r}.$$

Summary

- **Sequential:** sum of times
- **XOR:** weighted average
- **AND:** maximum branch time
- **Rework:** CT/(1-r) formula

Performance Metrics

- Cycle Time Efficiency = Theoretical CT / Actual CT
- Critical Path Method (CPM) identifies longest dependent sequence

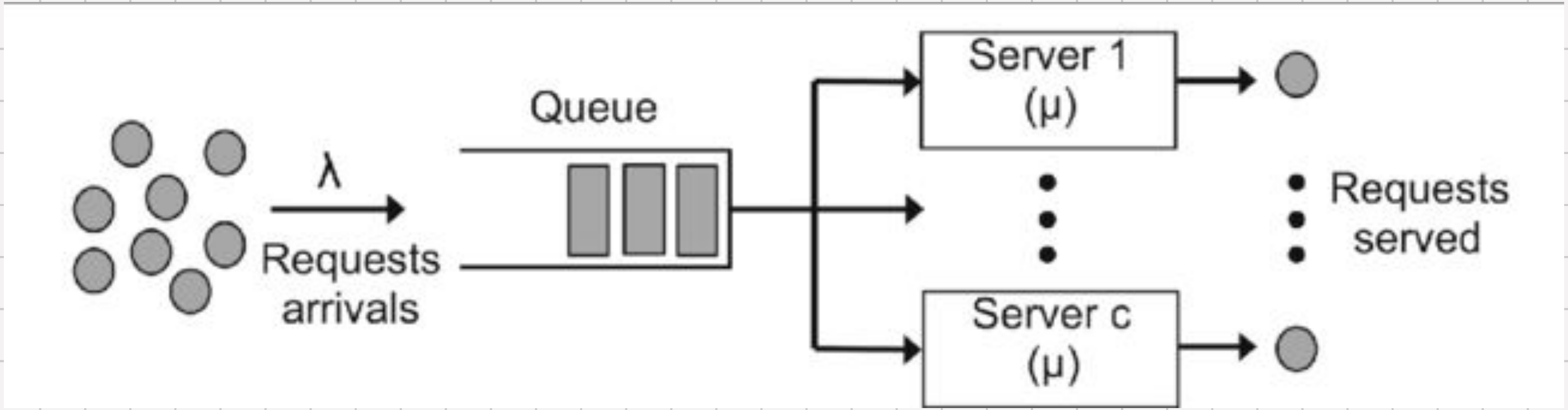
$$CTE = \frac{TCT}{CT}$$

Queueing Theory

Introduction to Queueing

- Explains resource contention and wait times
- Uses probability distributions for arrival and service rates

Basic Queue



λ
Requests arrivals
Queue
Server 1 (μ)
Server c (μ)
Requests served

Little's Law

An airport security checkpoint has:

L WIP (people in system) = 30 people

W Avg. time in system (CT) = 10 min = 1/6 hr

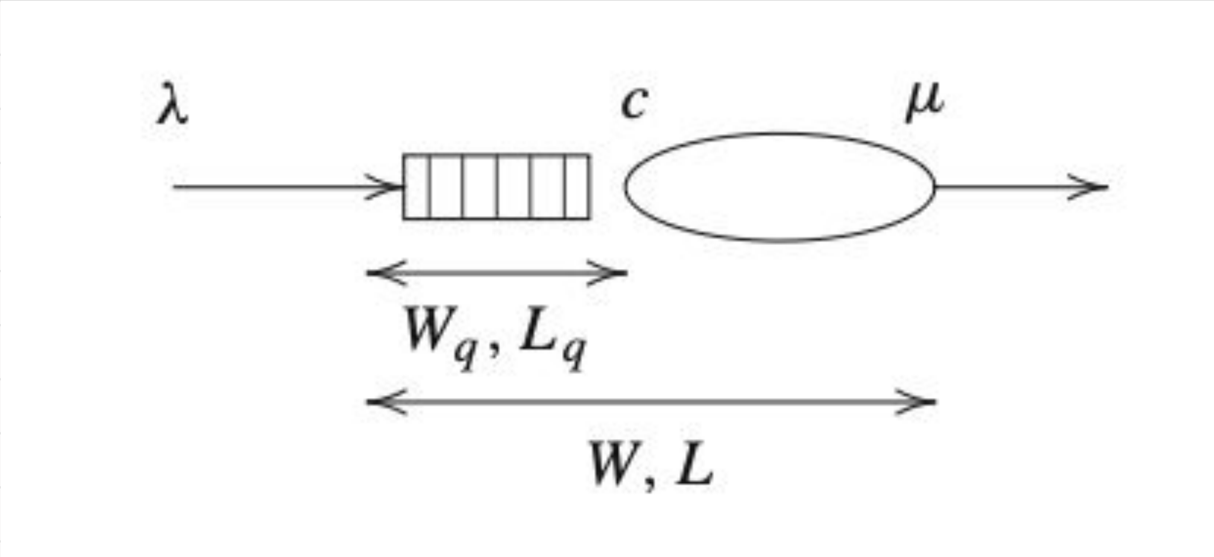
λ Arrival rate = ?

$$\lambda = L / W = 30 / (1/6)$$

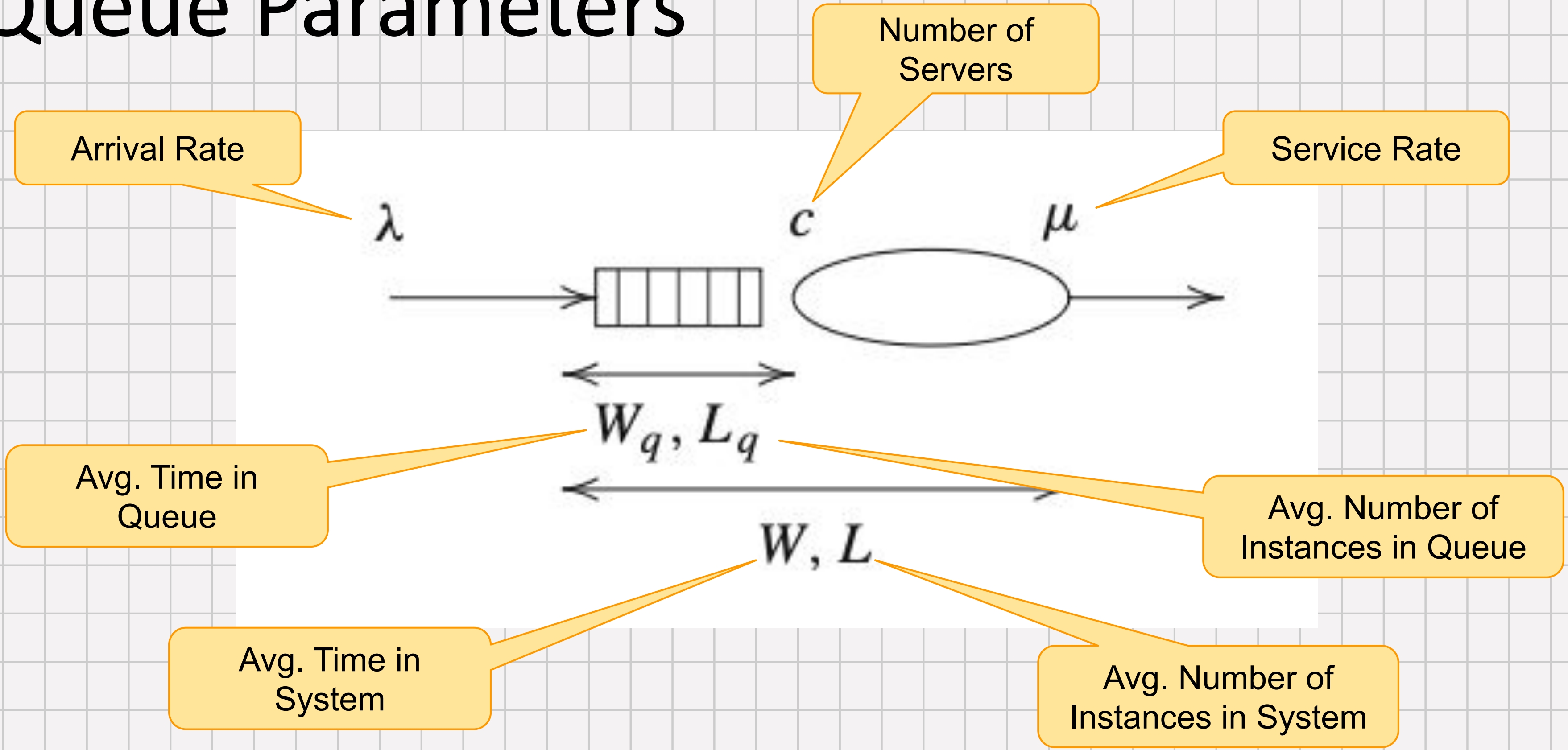
$\lambda = 180$ / hr

Know any two of L, W, λ — use Little's Law to find the third.

Queue Parameters



Queue Parameters



Example: M/M/1 Queue

Help desk receives 6 tickets/hr; agent resolves 10 tickets/hr (c = 1)

$\lambda = 6$ / hr (arrival rate)

$\mu = 10$ / hr (service rate)

$\rho = \lambda/\mu = 0.6$ (utilization)

Lq (queue length)	$\rho^2 / (1-\rho)$	**0.36 / 0.4 = 0.9**
Wq (wait in queue)	Lq / λ	**0.9 / 6 = 9 min**
W (total time)	Wq + $1/\mu$	**9 + 6 = 15 min**
L (system WIP)	$\lambda \times$ W	**6 × 0.25 = 1.5**

Avg. wait in queue = 9 min | Avg. total time = 15 min | Avg. WIP = 1.5 tickets

Generalization

M/M/1 Queue:

$$L_q = \rho^2/(1-\rho)$$

$$W_q = \frac{L_q}{\lambda}$$

$$W = W_q + \frac{1}{\mu}$$

$$L = \lambda W$$

M/M/c Queue:

$$L_q = \frac{(\lambda/\mu)^c \rho}{c!(1-\rho)^2 \left(\frac{(\lambda/\mu)^c}{c!(1-\rho)} + \sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} \right)}$$

Summary

- M/M/1 Queue: Single server, exponential interarrival and service times
- M/M/c Queue: Multiple servers handle arrivals in parallel
- **Limitations:** Focuses on isolated tasks rather than complex networks; assumes exponential distribution

Simulation

Purpose of Simulation

- Widely used for quantitative process analysis
- Replicates execution of many process instances

Simulation Inputs

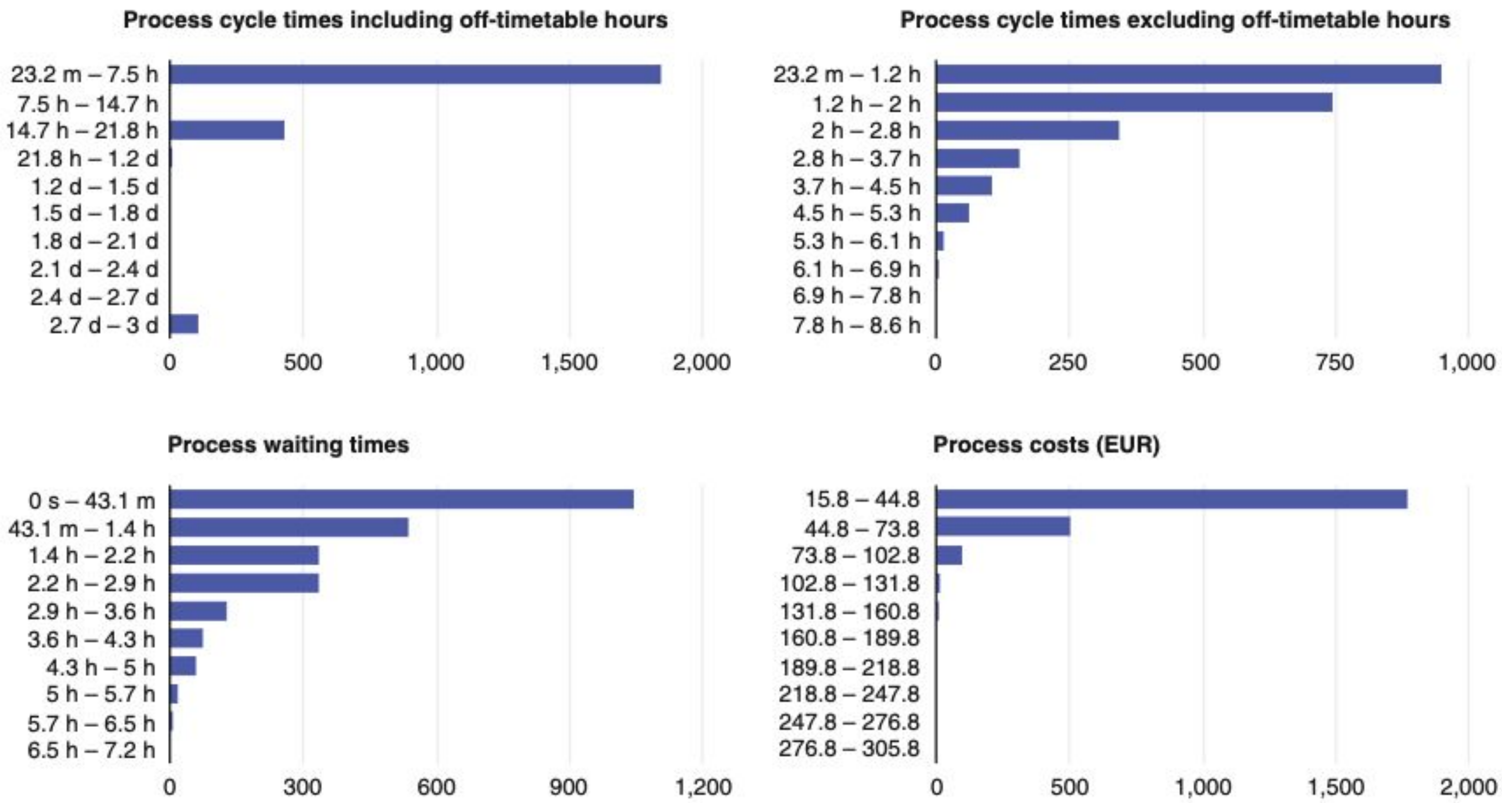
- **Distributions for tasks:** fixed, exponential, normal
- Branching probabilities, interarrival times

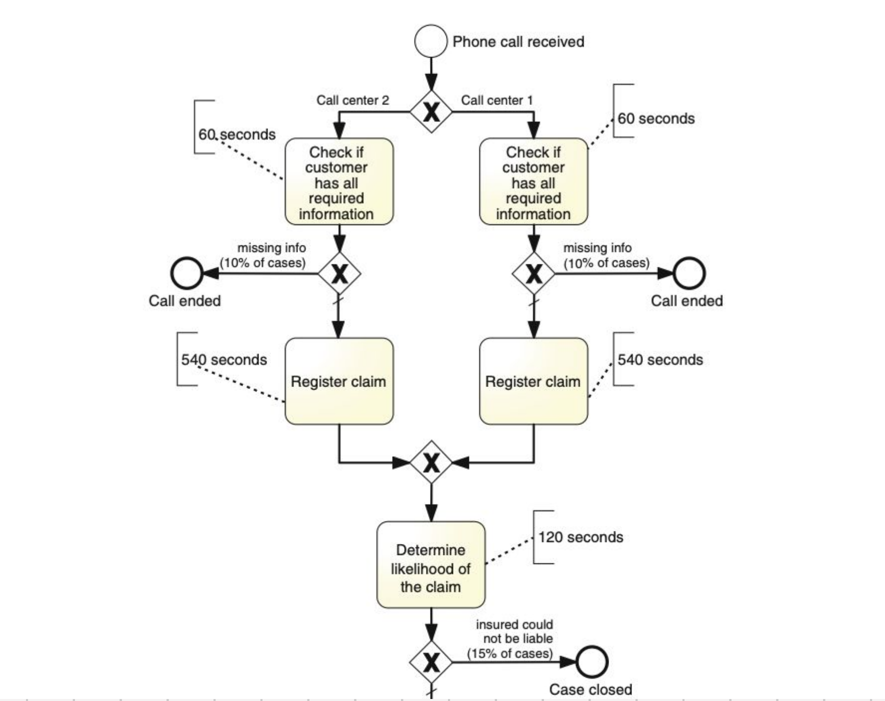
Simulation Outputs

- Cycle time distributions, waiting times, resource utilization
- Cost tracking and bottleneck identification

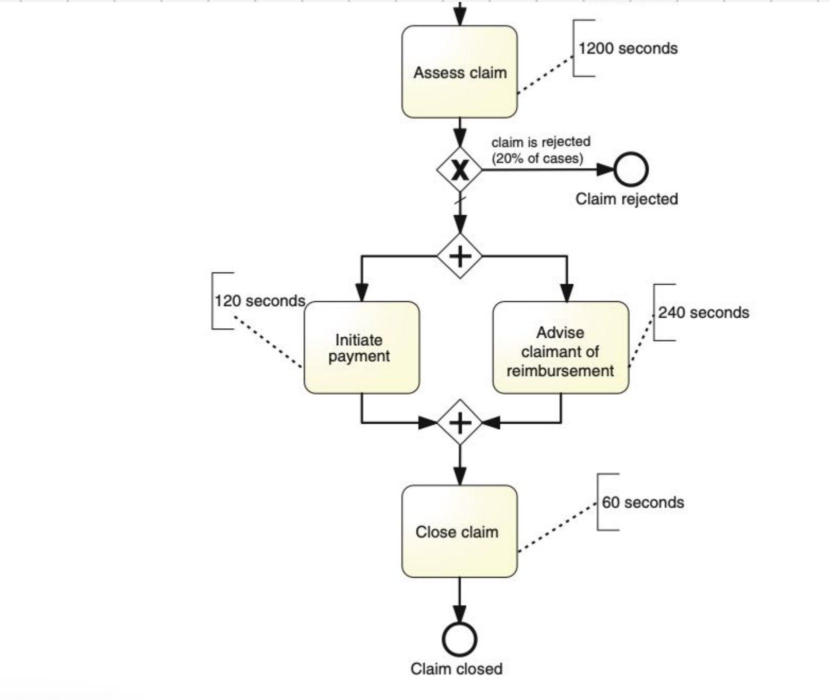
Advantages of Simulation

- Handles complexity beyond formulas
- Enables 'what-if' scenario testing and redesign exploration


Process cycle times including off-timetable hours
23.2 m – 7.5 h
7.5 h – 14.7 h
14.7 h – 21.8 h
21.8 h – 1.2 d
1.2 d – 1.5 d
1.5 d – 1.8 d
1.8 d – 2.1 d
2.1 d – 2.4 d
2.4 d – 2.7 d
2.7 d – 3 d
0
500
1,000
1,500
2,000
Process cycle times excluding off-timetable hours
23.2 m – 1.2 h
1.2 h – 2 h
2 h – 2.8 h
2.8 h – 3.7 h
3.7 h – 4.5 h
4.5 h – 5.3 h
5.3 h – 6.1 h
6.1 h – 6.9 h
6.9 h – 7.8 h
7.8 h – 8.6 h
0
250
500
750
1,000
Process waiting times
0 s – 43.1 m
43.1 m – 1.4 h
1.4 h – 2.2 h
2.2 h – 2.9 h
2.9 h – 3.6 h
3.6 h – 4.3 h
4.3 h – 5 h
5 h – 5.7 h
5.7 h – 6.5 h
6.5 h – 7.2 h
0
300
600
900
1,200
Process costs (EUR)
15.8 – 44.8
44.8 – 73.8
73.8 – 102.8
102.8 – 131.8
131.8 – 160.8
160.8 – 189.8
189.8 – 218.8
218.8 – 247.8
247.8 – 276.8
276.8 – 305.8
0
500
1,000
1,500
2,000



Phone call received
Call center 2
Call center 1
60 seconds
60 seconds
Check if customer has all required information
Check if customer has all required information
missing info (10% of cases)
missing info (10% of cases)
Call ended
Call ended
540 seconds
540 seconds
Register claim
Register claim
Determine likelihood of the claim
120 seconds
insured could not be liable (15% of cases)
Case closed



Assess claim
1200 seconds
claim is rejected (20% of cases)
Claim rejected
120 seconds
Initiate payment
Advise claimant of reimbursement
240 seconds
Close claim
60 seconds
Claim closed

Applications

Comparing Methods

- **Flow analysis:** structured formulas, averages
- **Queueing:** stochastic, includes contention effects
- **Simulation:** flexible, realistic, scalable

Practical Applications

- Predict cycle times and costs
- Identify bottlenecks and resource imbalances
- Support investment and redesign decisions